Pediatric Status Epilepticus

Section 1: Case Summary

Scenario Title:	Pediatric Status Epilepticus
Keywords:	Seizure, Peds, Status Epilepticus, Neurology
Brief Description of Case:	Pre-notification is sent about an 8-year-old with known seizure disorder coming in via EMS who has been seizing for 7 minutes and is persistently seizing despite intramuscular midazolam. The case will involve managing pediatric status epilepticus and including escalating anti-epileptics, intubation, and handing over to pediatrics.

Goals and Objectives	
Educational Goal:	Review management of pediatric status epilepticus
Objectives: (Medical and CRM)	1. Demonstrate an organized approach to the initial evaluation of a seizing pediatric patient 2. Call for help in a timely manner 3. Escalate anti-epileptics in a stepwise manner and consider intubation and sedation when appropriate 4. Consider possible etiologies of seizure 5. Communicate the clinical situation clearly to consultants and initiate transfer to a higher level of care
EPAs Assessed:	F1: Initiating and assisting in resuscitation of critically ill patients C1: Resuscitating and coordinating care for critically ill patients

Learners, Setting and Personnel				
Target Learners:	☒ Junior Learners		☒ Senior Learners	☐ Staff
	☒ Physicians	☐ Nurses	☐ RTs	☐ Inter-professional
	☐ Other Learners:			
Location:	☒ Sim Lab		☐ In Situ	☐ Other:
Recommended Number of Facilitators:	Instructors: 1			
	Confederates: 1 (parent)			
	Sim Techs: 1			

Scenario Development	
Date of Development:	June 14, 2021
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Last Revision Date:	
Revised By:	
Version Number:	

Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Nathan			Age: 8	Gender: M	Weight: 28kg
Presenting complaint: Seizure					
Temp: 37.6	HR: 125	BP: 95/60	RR: 24	O_2Sat: 97%	FiO_2: R/A
Cap glucose: 2.9 mmol/L			GCS: Actively seizing		
Triage note: You receive a pre-notification for an 8-year-old male with known seizure disorder who will arrive in 5 minutes. He has been seizing for roughly 7 minutes now and has received 1 dose of intramuscular midazolam 5mg.					
Allergies: None					
Past Medical History: Epilepsy Fully Vaccinated			Current Medications: Leviteracetam (recently stopped)		

Section 2B: Extra Patient Information

A. Further History
According to Mom, the patient has been feeling well leading up to today. There is no history of trauma, accidental ingestion, fever, or other symptoms suggestive of infection. The patient had been taking leviteracetam previously but hasn't had a seizure in 2 years, so Mom wanted to try him off medications and on a 'Keto' diet. He stopped taking his leviteracetam 2 weeks ago.

B. Physical Exam	
Cardio: Tachycardic, otherwise normal	Neuro: Generalized tonic-clonic seizure
Resp: Normal exam	Head & Neck: Patient has vomited, foam present at lips, not currently vomiting
Abdo: Normal	MSK/skin: Bilateral, convulsive muscle contractions
Other: N/A	

Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin *(Child)*
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Broselow tape and cart/drawer for 8-year-old (28kg) Airway equipment including laryngoscope, OPA, ET Tubes IV lines, NRB mask, Bag Valve Mask Monitors, EKG leads, Pulse Oximeter, Temp Probe
C. Required Medications
Midazolam/Lorazepam Phenytoin, Fosphenytoin, Levetiracetam, or Valproate Propofol, Rocuronium
D. Moulage
None
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed ☒ Patient not yet on monitor
F. Patient Reactions and Exam
Patient will be seizing throughout case and will continue to seize until intubated, sedated, and paralyzed with propofol and rocuronium respectively.

Section 4: Sim Actor and Standardized Patients

Sim Actor and Standardized Patient Roles and Scripts	
Role	*Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)*
Mom	Will present with patient and EMS. Can provide further background information, answer any further history questions, and highlight non-compliance with medications.
ED Nurse	Skillful and helpful. Provides support by applying monitors, starting IVs, administering medications, and updating team on patient's status.
RT	Assists with suctioning, airway support, and preparation with intubation.

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: Sinus Tach HR: 125 BP: 95/60 RR: 24 O_2SAT: 97% T: 37.6°C	Generalized tonic-clonic seizure	Expected Learner Actions ☐ Apply monitors and get vitals ☐ Check blood glucose (2.9 mmol/L) ☐ Call for help ☐ Apply O2 via NRB ☐ Obtain IV Access ☐ D10W 5cc/kg IV ☐ Midazolam 5mg (0.2mg/kg) IM or Lorazepam 1.5-2.5mg IV (0.05-0.1mg/kg) ☐ Draw up non-benzo anti-epileptic	Modifiers - Can include difficult IV access and move to IO Triggers *For progression to next state* - After 2nd dose of *Benzodiazepines* seizure continues	
2. Ongoing Seizure HR: 130 BP: 98/68 RR: 20 O2 Sat: 90% T: 37.5 °C	Patient continues to seize. Signs of airway compromise.	Expected Learner Actions ☐ Obtain further information from patient's mother ☐ Consider 3rd benzodiazepine dose ☐ Initiate anti-epileptics. One of: IV Fosphenytoin (20mg/kg over 10 mins), IV Levetiracetam (60mg/kg over 15 mins), IV Phenytoin 20mg/kg over 20 mins in NS), or IV Phenobarbital (20mg/kg) ☐ Repeat blood glucose (5.2mmol/L) ☐ Obtain labs (consider blood cultures) ☐ +/- Abx: Ceftriaxone and Vancomycin ☐ Consideration for CT Head	Modifiers - If asks for Peds/PICU, not readily available - Continues to seize, despite repeat benzos and second line agents Triggers *For progression to Intubation* - Increasing foam at mouth, patient begins to vomit - SpO2 slowly drops	
3. Intubation HR: 120 BP: 93/59 RR: 30	Unchanged	Expected Learner Actions ☐ Verbalize airway plan ☐ Use Broselow tape or clinical tool to select appropriate size tube, airway equipment and medications	Modifiers - Patient stops seizing with Rapid Sequence Intubation	

O2 Sat: 85% T: 37.4 ^{o}C		☐ Perform RSI ☐ Confirm tube placement ☐ Call neurology for consideration of continuous EEG monitoring	<u>Triggers</u> - Patient stabilizes after intubation -> PICU calls back	
4. Sedation and Handover	Patient paralyzed and sedated.	<u>Expected Learner Actions</u> ☐ Choose appropriate sedation ☐ Reassess ☐ Verbalize summary of case and plan to PICU	<u>Modifiers</u> - Prompt learner to present case for transport when PICU calls <u>Triggers</u> - Arrange transport-> End case	

Appendix A: Laboratory Results

CBC WBC 9 Hgb 128 Plt 210 Lytes Na 138 K 4.3 Cl 103 Urea 5 Cr 30 Glucose 5.2 Extended Lytes Ca 2.3 Mg 1.1 PO_4 1.5 Albumin TSH VBG pH 7.35 pCO_2 39 pO_2 90 HCO_3 21 Lactate 4.6	Cardiac/Coags INR 1.0 aPTT 30 Biliary AST 23 ALT 21 GGT 16 Bili 4 Lipase 17 Tox EtOH neg ASA neg Tylenol neg

Appendix B: ECGs, X-rays, Ultrasounds and Pictures

Paste in any auxiliary files required for running the session. Don't forget to include their source so you can find them later!

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

Questions for Discussions/Debrief

1. What is your differential for etiologies of pediatric seizure?
2. What are your initial priorities in the assessment and treatment of a seizing patient?
3. What is the definition of Status Epilepticus and why is it important?
4. What is your algorithm for the treatment of Pediatric Status Epilepticus?

References

1. https://www.cps.ca/en/documents/position/emergency-management-of-the-paediatric-patient-with-convulsive-status-epilepticus
2. https://trekk.ca/resources?tag_id=D013226
3. https://emergencymedicinecases.com/emergency-management-of-pediatric-seizures/
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